

PIYUSH KUMAR SINGH

AI/ML ENGINEER | MACHINE LEARNING | DEEP LEARNING | GENERATIVE AI | PYTHON

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [CodeChef](#)

SUMMARY

B.Tech Computer Science student with hands-on experience in Python, machine learning, deep learning, computer vision, and full-stack development. Built and evaluated AI/ML systems using TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, Pandas, and NumPy, and deployed models through FastAPI with MongoDB-backed web applications. Demonstrated problem-solving through 500+ coding problems and production-oriented projects covering fraud detection, defect classification, and MERN application development.

EXPERIENCE

OpenPaymnt | Software Development Intern

Jul 2025 - Dec 2025

- Implemented responsive frontend features with React.js and Next.js, including API integration, debugging, and UI performance improvements.
- Supported backend development tasks involving MongoDB and AI/ML workflows while collaborating with the product team through Git and GitHub.

PROJECTS

Fraud & Risk Detection System

Python | Scikit-learn | FastAPI | MongoDB | Real-time ML [GitHub](#)

- Engineered an end-to-end machine learning pipeline for real-time fraud and high-risk transaction detection using a financial transaction dataset.
- Built preprocessing and feature-engineering pipelines and evaluated classification models to achieve 87% F1-score and 91% fraud recall on the test set.
- Deployed the trained model with FastAPI and MongoDB and developed a web interface for transaction input, real-time predictions, and prediction history.

Defect Detection System

Python | TensorFlow | Keras | OpenCV | Grad-CAM [GitHub](#)

- Developed a CNN-based manufacturing defect classifier with transfer learning across 15 defect categories, reaching 91.17% test accuracy and 89.37% balanced accuracy.
- Built image preprocessing and training workflows for 5,354 images using resizing, normalization, and augmentation to improve model generalization.
- Integrated Grad-CAM visualizations to identify image regions influencing model predictions and support model interpretability.

Job Portal Application

MongoDB | Express.js | React.js | Node.js [GitHub](#)

- Developed a full-stack MERN job portal with authentication, job posting, search and filtering, applications, and resume uploads for job seekers and recruiters.
- Designed REST-based backend workflows and responsive React interfaces for recruiter-side application management and applicant-side job discovery.

TECHNICAL SKILLS

Programming Languages: Python, C++

Machine Learning: Supervised Learning, EDA, Feature Engineering, Model Evaluation, Scikit-learn, XGBoost, Gradient Boosting, Hyperparameter Tuning

Deep Learning & Computer Vision: TensorFlow, Keras, CNNs, Transfer Learning, OpenCV, Grad-CAM, Image Augmentation

Frontend & Backend Development: React.js, Next.js, Node.js, Express.js, MongoDB, FastAPI, REST APIs

Core CS: Data Structures & Algorithms, Computer Networks

Tools & Technologies: Git, GitHub, Postman, VS Code

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Introduction to Data Science - Cisco Networking Academy; Data Analytics Job Simulation - Deloitte; Getting Started with Generative AI - IBM; Make Agentic AI - IBM; AI Foundation; Java Programming Certification - IIT Bombay; Python (Basic) Certification - HackerRank.

Achievements: 500+ problems solved on LeetCode | 3-star rating on CodeChef as a competitive programmer

EDUCATION

ITM Skills University, Navi Mumbai | Bachelor of Technology in Computer Science

2023 - 2027 (Expected) | CGPA: 8.6 / 10.0

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems, Machine Learning, Probability and Statistics